



AZITHROMYCIN 500MG

I.V. FOR INFUSION

COMPOSITION:

Each vial contains Azithromycin 500mg

PRESENTATION:

A white or almost white hygroscopic crystalline powder that provides a clear solution after reconstitution.

INDICATIONS:

Azithromycin intravenous (IV) is indicated for the treatment of community acquired pneumonia (CAP) caused by susceptible organisms, including *Legionella pneumophila*, in patients who require initial intravenous therapy.

Azithromycin intravenous (IV) is indicated for the treatment of pelvic inflammatory disease (PID) caused by susceptible organisms (*Chlamydia trachomatis*, *Neisseria gonorrhoea*, *Mycoplasma hominis*), in patients who require initial intravenous therapy.

PHARMACODYNAMICS:

Azithromycin is the first of a subclass of macrolides antibiotics, known as azalides, and is chemically different from erythromycin. Chemically it is derived by insertion of a nitrogen atom into the lactone ring of erythromycin A. The mode of action of azithromycin is inhibition of protein synthesis in bacteria by binding to the 50s ribosomal subunit and preventing translocation of peptides. Azithromycin demonstrates activity in vitro against a wide range of bacteria including:

Gram-positive Aerobic Bacteria – *Staphylococcus aureus*, *Streptococcus pyogenes* (group A beta – hemolytic streptococci), *Streptococcus pneumoniae*, alpha – hemolytic streptococci (viridans group) and other streptococci, and *Corynebacterium diphtheriae*. Azithromycin demonstrates cross-resistance with erythromycin-resistant Gram-positive strains, including *Streptococcus faecalis* (enterococcus) and most strains of methicillin-resistant staphylococci.

Gram-negative Aerobic Bacteria – *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Moraxella catarrhalis*, *Acinetobacter* species, *Yersinia* species, *Legionella pneumophila*, *Bordetella pertussis*, *Bordetella parapertussis*, *Shigella* species, *Pasteurella* species, *Vibrio cholerae* and *parahaemolyticus*, *Plesiomonas shigelloides*. Activities against *Escherichia coli*, *Salmonella enteritidis*, *Salmonella typhi*, *Enterobacter* species, *Aeromonas hydrophila* and *Klebsiella* species are variable and susceptibility tests should be performed. *Proteus* species, *Serratia* species, *Morganella* species, and *Pseudomonas aeruginosa* are usually resistant.

Anaerobic Bacteria – *Bacteroides fragilis* and *Bacteroides* species, *Clostridium perfringens*, *Peptococcus* species and *Peptostreptococcus* species, *Fusobacterium necrophorum* and *Propionibacterium acnes*.

Organisms of Sexually Transmitted Diseases - Azithromycin is active against *Chlamydia trachomatis* and also shows good activity against *Treponema pallidum*, *Neisseria gonorrhoeae*, and *Haemophilus ducreyi*.

Other Organisms – *Borrelia burgdorferi* (Lyme disease agent), *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, *Mycoplasma hominis*, *Ureaplasma urealyticum*, *Campylobacter* species and *Listeria monocytogenes*.

Opportunistic Pathogens Associated with HIV Infections – *Mycobacterium avium-intracellulare* complex, *Pneumocystis carinii* and *Toxoplasma gondii*.

Commonly susceptible species:

Aerobic Gram-positive bacteria:

Staphylococcus aureus, *Streptococcus agalactiae*, Streptococci (Groups C, F, G) and Viridans group streptococci.

Aerobic Gram-negative bacteria:

Bordetella pertussis, *Haemophilus ducreyi*, *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Legionella pneumophila*, *Moraxella catarrhalis* and *Neisseria gonorrhoeae*.

Other:

Chlamydia pneumoniae, *Chlamydia trachomatis*, *Mycoplasma pneumoniae* and *Ureaplasma urealyticum*.

Species for which acquired resistance has been reported:

Aerobic Gram-positive bacteria:

Streptococcus pneumoniae

Streptococcus pyogenes

Note: Azithromycin demonstrates cross-resistance with erythromycin-resistant gram-positive strains.

Inherently resistant organisms:

Enterobacteriaceae

Pseudomonas

PHARMACOKINETICS:

Absorption:

Following oral administration in humans, azithromycin is widely distributed throughout the body; bioavailability is approximately 37%. The time taken to peak plasma levels is 2-3 hours.

Distribution:

In animal studies, high azithromycin concentrations have been observed in phagocytes. In experimental models, higher concentrations of azithromycin are released during active phagocytosis than from non-stimulated phagocytes. In animal models this results in high concentrations of azithromycin being delivered to the site of infection. Pharmacokinetic studies in humans have shown markedly higher azithromycin levels in tissue than in plasma (up to 50 times the maximum observed concentration in plasma) indicating that the drug is heavily tissue bound. Concentrations in target tissues, such as lung, tonsil and prostate exceed the MIC₉₀ for likely pathogens after a single dose of 500mg.

Elimination:

Plasma terminal elimination half-life closely reflects the tissue depletion half-life of 2 to 4 days. Approximately 12% of an intravenously administered dose is excreted in the urine over 3 days as the parent drug, the majority in the first 24 hours. Biliary excretion of azithromycin is a major route of elimination for unchanged drug following oral administration. Very high concentrations of unchanged drug have been found in human bile, together with 10 metabolites, formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings, and by cleavage of the cladinose conjugate. Comparison of HPLC and microbiological assays in tissues suggests that metabolites play no part in the microbiological activity of azithromycin.

Pharmacokinetics in Special Patient Groups:

Elderly:

In elderly volunteers (>65 years), slightly higher AUC values were seen after a 5-day regimen than in young volunteers (<40 years), but these are not considered clinically significant, and hence no dose adjustment is recommended.

Renal Impairment:

The pharmacokinetics of azithromycin in subjects with mild to moderate renal impairment (GFR10-80ml/min) were not affected following a single

one gram dose of immediate release azithromycin. Statistically significant differences in AUC₀₋₁₂₀ (8.8µg·hr/ml vs. 11.7µg·hr/ml), C_{max} (1.0µg/ml vs. 1.6µg/ml) and CL_r 2.3ml/min/kg vs. 0.2ml/min/kg) were observed between the group with severe renal impairment (GFR<10ml/min) and the group with normal renal function.

Hepatic Impairment:

In patients with mild to moderate hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to those with normal hepatic function. In these patients urinary clearance of azithromycin appears to increase, perhaps to compensate for reduced hepatic clearance.

DOSAGE AND ADMINISTRATION:

Adults:

For the treatment of adult patients with CAP due to the indicated organisms, the recommended dose of intravenous azithromycin is 500mg as a single dose by the IV route for one or two days. Intravenous therapy should be followed by oral azithromycin as a single daily dose of 500mg to complete a 7 to 10 day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response. For the treatment of adult patients with PID due to the indicated organisms, the recommended dose of intravenous azithromycin is 500mg as a single dose by the IV route for one or two days. Intravenous therapy should be followed by azithromycin by the oral route at a single daily dose of 250mg to complete a 7-day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response. If anaerobic microorganisms are suspected of contributing to the infection, an antimicrobial anaerobic agent may be administered in combination with azithromycin.

Children:

The safety and efficacy of intravenous azithromycin for the treatment of infections in children has not been established.

Elderly: The same dosage as in adult patients is used in the elderly.

Patients with Renal Impairment:

No dosage adjustment is necessary in patients with mild to moderate renal impairment (GFR10 – 80ml/min). Caution should be exercised when azithromycin is administered to patients with severe renal impairment (GFR < 10ml/min).

Patients with Hepatic Impairment:

The same dosage as in patients with normal hepatic function may be used in patients with mild to moderate hepatic impairment.

ROUTE OF ADMINISTRATION:

Intravenous Administration:

After reconstitution and dilution, the recommended route of administration for intravenous azithromycin is **by IV infusion only. Do not administer as an intravenous bolus or an intramuscular injection.** The infusate concentration and rate of infusion for azithromycin intravenous (IV) should be either 1 mg/ml over 3 hours or 2mg/ml over 1 hour. An intravenous dose of 500mg azithromycin should be infused for a minimum duration of one (1) hour.

Intravenous (IV) solution preparation:

Reconstitution:

Prepare the initial IV solution for infusion by adding 4.8ml of sterilized Water for Injection to the vial and shake the vial until all of the drug is dissolved. Each ml of reconstituted solution contains 100mg azithromycin. Chemical and physical in-use stability of the reconstituted product has been demonstrated for 24 hours at 25±2°C. When diluted according to the instructions the diluted solution is chemically and physically stable for 24 hours at or below 25°C or for 7 days if stored under refrigeration 2-8°C. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally be no longer than 24 hours at 2 to 8°C, unless reconstitution and dilution have taken place in controlled and validated aseptic conditions.

Dilute this solution further prior to administration as instructed below:

Dilution:

To provide azithromycin over a concentration range of 1.0 - 2.0mg/ml, transfer 5ml of the 100mg/ml azithromycin solution into the appropriate amount of any of the diluents listed below :

Final Infusion Solution Concentration (mg/ml)	Amount of Diluents (ml)
1.0mg/ml	500ml
2.0mg/ml	250ml

The reconstituted solution can be diluted with:

- Normal Saline (0.9% sodium chloride)
- 1/2 Normal Saline (0.45% sodium chloride)
- 5% Dextrose in Water
- Lactated Ringer's Solution
- 5% Dextrose in 1/2 Normal Saline (0.45% sodium chloride) with 20mEq KCl
- 5% Dextrose in Lactated Ringer's Solution
- 5% Dextrose in 1/3 Normal Saline (0.3% sodium chloride)
- 5% Dextrose in 1/2 Normal Saline (0.45% sodium chloride)
- Normosol® - M in 5% Dextrose
- Normosol® - R in 5% Dextrose

Parenteral drug products should be inspected visually for particulate matter prior to administration. If particulate matter is evident in reconstituted fluids, the drug solution should be discarded.

CONTRAINDICATIONS:

The use of this product is contraindicated in patients with hypersensitivity to azithromycin, erythromycin, any macrolide or ketolide antibiotic, or to any excipient.

SIDE EFFECTS:

Azithromycin is well tolerated with a low incidence of side effects.

Blood and Lymphatic System Disorders: Transient episodes of mild neutropenia have occasionally been observed, although a causal relationship to azithromycin has not been established.

Ear and Labyrinth Disorders: Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported due to prolonged use of high doses in investigational studies. In those cases where follow-up information was available the majority of these events were reversible.

Gastrointestinal Disorders: Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps) and flatulence.

Hepatobiliary Disorders: Abnormal liver function.

Skin and Subcutaneous Tissue Disorders: Allergic reactions including rash and angioedema.

General Disorders and Administration Site Conditions: Local pain and inflammation at the site of infusion.

The following undesirable effects have been reported in association with DMAC prophylaxis and treatment: The most frequent adverse reactions in HIV-infected patients receiving azithromycin for prophylaxis for DMAC were diarrhea, abdominal pain, nausea, loose stools, flatulence, vomiting, dyspepsia, rash, pruritus, headache and arthralgia. When azithromycin 600mg is given daily for the treatment of DMAC infection for prolonged periods, the most frequently reported treatment-related side effects are abdominal pain, nausea, vomiting, diarrhea, flatulence, headache, abnormal vision, and hearing impairment.

In post-marketing experience, the following additional undesirable effects have been reported:

Infections and Infestations: Moniliasis and vaginitis

Blood and Lymphatic System Disorders: Thrombocytopenia

Immune System Disorders: Anaphylaxis (rarely fatal)

Metabolism and Nutrition Disorders: Anorexia

Psychiatric Disorders: Aggressive reaction, nervousness, agitation and anxiety.

Nervous System Disorders: Dizziness, convulsions (as seen with other macrolides), headache, hyperactivity, hypoesthesia, paresthesia, somnolence, and syncope. There have been rare reports of taste / smell perversion and / or loss. However, a causal relationship has not been established.

Ear and Labyrinth Disorders: Vertigo

Cardiac Disorders: Palpitations and arrhythmias including ventricular tachycardia (as seen with other macrolides) have been reported. There have been rare reports of QT prolongation and torsades de pointes. A causal relationship between azithromycin and these effects has not been established.

Vascular Disorders: Hypotension

Gastrointestinal Disorders: Vomiting / diarrhea (rarely resulting in dehydration), dyspepsia, constipation, pseudomembranous colitis, pancreatitis, rare reports of tongue discoloration and infantile hypertrophic pyloric stenosis.

Hepatobiliary Disorders: Hepatitis and cholestatic jaundice have been reported, as well as rare cases of hepatic necrosis and hepatic failure, which have rarely resulted in death. However, a causal relationship has not been established.

Skin and Subcutaneous Tissue Disorders: Allergic reactions including pruritus, rash, photosensitivity, edema, urticaria, and angioedema. Rarely, serious cutaneous adverse reactions including erythema multiforme, SJS, TEN and DRESS have been reported.

Musculoskeletal and Connective Tissue Disorders: Arthralgia

Renal and Urinary Disorders: Interstitial nephritis and acute renal failure.

General Disorders and Administration Site Conditions: Asthenia has been reported, although a causal relationship has not been established, fatigue and malaise.

SYMPTOMS AND TREATMENTS FOR OVERDOSAGE:

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. In the event of overdose, general symptomatic and supportive measures are indicated as required.

WARNING AND PRECAUTION:

As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment. Since liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease. In patients receiving ergot derivatives, ergotism has been precipitated by coadministration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be co-administered. As with any antibiotic preparation, observation for signs of superinfection with non-susceptible organisms, including fungi is recommended. *Clostridium difficile* associated diarrhea (CDAD) has been reported with use of nearly all antibacterial agents, including azithromycin, and may range in severity from mild diarrhea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*. *C. difficile* produces toxins A and B which contribute to the development of CDAD. Hypertoxic producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhea following antibiotic use. Careful medical history is necessary since CDAD has been reported

to occur over two months after the administration of antibacterial agents. In patients with severe renal impairment (GFR<10ml/min) a 33% increase in systemic exposure to azithromycin was observed. Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with other macrolides. A similar effect with azithromycin cannot be completely ruled out in patients at increased risk for prolonged cardiac repolarization.

Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and / or irritability with feeding occurs.

Intravenous Administration:
Azithromycin for injection should be reconstituted and diluted as directed and administered as an intravenous infusion over not less than 60 minutes. **Do not administer as an intravenous bolus or an intramuscular injection.**

Hypersensitivity:
As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), dermatologic reactions including Stevens-Johnson Syndrome (SJS), Toxic Epidermal Necrolysis (TEN) (rarely fatal), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment. If an allergic reaction occurs, the drug should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

Prolongation of the QT interval:
Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:
- Patients with congenital or documented QT prolongation
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
- Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval

Use in Pregnancy:
Animal reproduction studies have been performed at doses up to moderately maternally toxic dose concentrations. In these studies, no evidence of harm to the fetus due to azithromycin was found. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, azithromycin should be used during pregnancy only if clearly needed.

Use in Lactation:
There are no data on secretion in breast milk. As many drugs are excreted in human milk, azithromycin should not be used in the treatment of a lactating woman unless the physician feels that the potential benefits justify the potential risks to the infant.

Effects on ability to drive and use machines:
There is no evidence to suggest that azithromycin may have an effect on the patient's ability to drive or operate machinery.

INTERACTION WITH OTHER MEDICAMENTS:
Antacids: No effect on overall bioavailability was seen although peak serum concentrations were reduced by approximately 25%. In patients receiving both oral azithromycin and antacids, the drugs should not be taken simultaneously.

Cetirizine: Coadministration of a 5-day regimen of azithromycin with cetirizine 20mg at steady-state resulted in no pharmacokinetic interaction and no significant changes in the QT interval.

Didanosine (Dideoxyinosine): Coadministration of 1200mg/day azithromycin with 400mg/day didanosine in 6 HIV-positive subjects did not appear to affect the steady-state pharmacokinetics of didanosine as compared with placebo.

Digoxin: Some of the macrolide antibiotics have been reported to impair the microbial metabolism of digoxin in the gut in some patients. In patients receiving concomitant azithromycin, a related azalide antibiotic, and digoxin the possibility of raised digoxin levels should be borne in mind.

Zidovudine: Single 1000mg doses and multiple 1200mg or 600mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding is unclear, but it may be of benefit to patients. Azithromycin does not interact significantly with the hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot: Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended. Pharmacokinetic studies have been conducted between azithromycin and the following drugs known to undergo significant cytochrome P450 mediated metabolism.

Atorvastatin: Coadministration of atorvastatin (10mg daily) and azithromycin (500mg daily) did not alter the plasma concentrations of atorvastatin.

Carbamazepine: No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine: In a pharmacokinetic study investigating the effects of a single dose of cimetidine, given 2 hours before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants: There have been reports of potentiated anticoagulation subsequent to coadministration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin: Caution should be exercised before considering concurrent administration of cyclosporin and azithromycin. If coadministration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz: Coadministration of a 600mg single dose of azithromycin and 400mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole: Coadministration of a single dose of 1200mg azithromycin did not alter the pharmacokinetics of a single dose of 800mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the coadministration of fluconazole, however a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir: Coadministration of a single dose of 1200mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800mg three times daily for 5 days.

Methylprednisolone: Azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam: Coadministration of azithromycin 500mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15mg dose of midazolam.

Nelfinavir: Coadministration of azithromycin (1200mg) and nelfinavir at steady state (750mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin: Coadministration of azithromycin and rifabutin did not affect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil: There was no evidence of an effect of azithromycin (500mg daily for 3 days) on the AUC and C_{max}, of sildenafil or its major circulating metabolite.

Terfenadine: Pharmacokinetic studies have reported no evidence of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline: There is no evidence of a clinically significant pharmacokinetic interaction when azithromycin and theophylline are co-administered to healthy volunteers.

Triazolam: Coadministration of azithromycin with 0.125mg triazolam had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim / sulfamethoxazole: Coadministration of trimethoprim / sulfamethoxazole DS (160mg / 800mg) for 7 days with azithromycin 1200mg on Day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other studies.

INCOMPATIBILITIES:

Intravenous: Other intravenous substances, additives or medications should not be added to intravenous azithromycin, or infused simultaneously through the same intravenous line.

STORAGE:

Store below 30°C. Protect from light. For storage after reconstitution, please refer section **Route of Administration**.

**KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK**

PACK QUANTITIES:

Available in 1 vial per box & 10 vials per box.
Not all pack sizes may be marketed.

Further information can be obtained from pharmacist, physician or the manufacturer.

Manufactured By & Product Holder:
 **Kotra Pharma (M) Sdn. Bhd.**
No. 1, 2 & 3, Jalan TTC 12, (90082-V)
Cheng Industrial Estate,
75250 Melaka, Malaysia.